MILITARY SPECIFICATION

DIHYDROERGOCORNINE, DIHYDROERGOCRISTINE, AND DIHYDROERGOKRYPTINE MESYLATES TABLETS

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers Dihydroergocornine, Dihydroergocristine, and Dihydroergokryptine Mesylates Tablets (see 6.2).

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-C-186 - Containers, Packaging and Packing for Drugs, Chemicals and Pharmaceuticals.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Hq, Defense Personnel Support Center, ATTN: Directorate of Medical Materiel, DPSC-ATT, 2800 South 20th Street, Philadelphia, PA 19101, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

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STANDARDS

FEDERAL

FED-STD-140 - Tablets (for Medicinal Purposes).

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitiation for bids or request for proposals shall apply.

U. S. DEPARTMENT OF HEALTH, EDUCATION, AND WELFARE,
FOOD AND DRUG ADMINISTRATION (FDA)

Federal Food, Drug, and Cosmetic Act and Regulations Promulgated Thereunder.

(Application for copies should be addressed to the Food and Drug Administration, Washington DC 20204.)

U. S. PHARMACOPEIAL CONVENTION, INC.

The United States Pharmacopeia (USP).

(Application for copies should be addressed to the Mack Publishing Company, Easton, PA 18042.)

The National Formulary (NF).

(Application for copies should be addressed to the Mack Publishing Company, Easton, PA 18042.)

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3. REQUIREMENTS

3.1 Material. Shall be Dihydroergocornine, Dihydroergocristine, and Dihydroergokryptine Mesylates Tablets and shall be in accordance with all applicable requirements of FED-STD-140, and as specified herein.

S2. Classification. Shall be type VI.

Shall be suitable for sublingual use.

S5.1 The active ingredient shall comply with the tests, standards and requirements as specified in 4.2.2.

S5.3 The following is added to this paragraph:

Shall be tablets containing 0.333 mg of Dihydroergocornine Mesylate, 0.333 mg of Dihydroergocristine Mesylate, and 0.333 mg of Dihydroergokryptine Mesylate per tablet in an equilibrium mixture, composed of approximately equal parts of each salt. The total alkaloidal content per tablet shall be within the designated assay limits for the tablets.

The ratio of the three alkaloidal salts shall be determined from the Quality Control Certificate of Analysis of the active ingredients as submitted by the manufacturer.

The tablets shall assay to contain not less than 90.0 percent and not more than 110.0 percent of the required amount of the mixture of alkaloidal salts when assayed as specified in 4.2.1.

S6.4.9.1 Disintegration. Tablets shall disintegrate in not more than 10 minutes when determined in accordance with the procedure described in the USP for "Sublingual Tablets."

3.2 Ingredients. All ingredients used in the manufacture of the tablets shall comply with S5.1 of FED-STD-140.

3.3 Expiration dating period. The expiration dating period shall be not less than 48 months.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Supplier responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and, (i) as to any expiration dated item, for the expiration dating period specified by the contractor for such item, or such longer period as may be required by regulation of any federal agency; or (ii) as to non-dated items, for not less than 3 years after delivery of the item to the Governemt.

4.1.2 No company supplying any ingredient(s) to the contractor will be considered an acceptable facility for the performance of any inspection requirements specified herein.

4.2 Tests.

4.2.1 Assay of alkaloid salts in the finished tablet.

Reagents:

1:10 Benzalkonium Chloride Solution. Dilute 10.0 ml of a 17.0 percent Benzalkonium Chloride solution to 17.0 ml with purified water.

p-Dimethylaminobenzaldehyde Reagent. Very carefully, with stirring, add 650 ml of Sulfuric Acid to 350 ml of purified water and cool the solution to room temperature. (DO NOT DILUTE THE SOLUTION TO 1000 ml.) Add 1.25 grams of p-Dimethylaminobenzaldehyde and 2 ml of freshly prepared 5 percent Ferric Chloride solution, and mix until solution is complete. (Do not expose this solution to light for prolonged periods, and refrigerate it when not in use.)

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Dihydroergocornine, Dihydroergocristine, and Dihydroergokrptine Mesylates Standard Solution (0.05 mg/ml).

NOTE: Use low-actinic flasks.

Dissolve approximately 50 mg of the Alkaloid Salts Reference Standard powder accurately weighed and previously dried to constant weight in vacuum at 80° C, in sufficient one (1) percent Tartaric Acid solution to make 100.0 ml. Dilute 10.0 ml of this solution to 100.0 ml with one (1) percent Tartaric Acid Solution.

Procedure: (Protect all solution from light.)

Place 10 tablets in a 200 ml volumetric flask. Add 150 ml of one (1) percent Tartaric Acid Solution and 4 drops of a 1:10 Benzalkonium Chloride Solution and shake on a mechanical shaker for 30 minutes. Adjust the volume to 200.0 ml with one (1) percent Tartaric Acid solution, mix well, and filter through a "fine" fritted glass funnel. Into separate Erlenmeyer flasks, pipet 5.0 ml of the filtrate, 5.0 ml of the Standard Solution, and 5.0 ml of one (1) percent Tartaric Acid solution for a blank. Add 10.0 ml of p-Dimethylaminobenzaldehyde Reagent to each flask, shake well, and allow to stand for 15 minutes in normal light. Determine the absorbances of the sample and standard solutions against the blank at 585 mu, with a suitable spectrophotometer, in a 1 cm cell.

Calculation:

$$\frac{A_{585\ \text{Sample}}}{A_{585\ \text{Standard}}} \times 1 = \text{mg of mixture of 3 alkaloid salts, per tablet.}$$

Where: $A_{585\ \text{Sample}}$ = absorbance of sample solution
$A_{585\ \text{Standard}}$ = absorbance of standard solution

(NOTE: Adjust calculation based on exact weight of standard.)

4.2.2 Active ingredient. The active ingredient shall be mixture of approximately equal parts of Dihydroergocornine Mesylate, Dihydroergocristine Mesylate and Dihydroergokryptine Mesylate, and shall comply with the following:

Identification. Dissolve a small amount of powder in 5 ml of purified water, add 5 ml of a mixture of equal volumes of glacial acetic acid and sulfuric acid, and add a drop of freshly prepared 5 percent ferric chloride solution. A violet-blue color shall develop.

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Loss on drying. Shall lose not more than 6.0 percent of its weight when dried to constant weight at 80° C, under vacuum, using the USP method. (Save dried sample for assay.)

Assay. Shall contain not less than 97.0 percent and not nore than 103.0 percent of a mixture of total alkaloid salts, when assayed as follows:

Reagents:

p-Dimethylaminobenzaldehyde Reagent (see 4.2.1).

Dihydroergocornine, Dihydroergocristine, and Dihydroergokryptine Mesylates Standard Solution (see 4.2.1).

Procedure:

Accurately weigh approximately 50 mg of dried sample and dissolve it in sufficient one (1) percent Tartaric Acid solution to make 100.0 ml. Dilute 10.0 ml of this solution to 100.0 ml with one (1) percent Tartaric Acid solution. Into separate Erlenmeyer flasks, pipet 5.0 ml of the final sample solution, 5.0 ml of the Standard Solution, and 5.0 ml of one (1) percent Tartaric Acid solution for a blank. Add 10.0 ml of p-Dimethylaminobenzaldehyde Reagent to each flask, shake well and allow to stand for 15 minutes in normal light. Determine the absorbance of the sample and standard solution against the blank at 585 mu, with a spectrophotometer, in a 1 cm cell.

Calculation: $\frac{A_{585\ \text{sample}}}{A_{585\ \text{standard}}} \times \frac{50}{W} \times 100 = \text{percent of alkaloid salts}$

Where $A_{585\ \text{sample}}$ = absorbance of sample solution
$A_{585\ \text{standard}}$ = absorbance of standard solution
W = sample weight in mg
50 = dilution factor = $(100 \times \frac{100}{10} \times 0.05)$

NOTE: Adjust dilution factor to reflect exact weight of standard.

The ratio of the three alkaloidal salts shall be determined from the Certificate of Analysis of the active ingredients as submitted by the manufacturer.

4.3 Packaging inspection. The inspection of the intermediate package, packing and marking for shipment and storage shall be in accordance with quality assurance provisions of the applicable container specification and marking requirements of MIL-STD-129.

4.4 Labeling inspection. Sampling shall be in accordance with MIL-STD-105 and the unit of product shall be the unit of issue. The inspection lot shall be the number of units of issue of each batch, lot, control number, etc., offered for inspection at one time. The inspection level for labeling shall be S-2 and the acceptance number shall be zero.

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5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with all applicable requirements of PPP-C-186, and as specified herein.

5.2 Packaging. Packaging shall be level A or C, as specified (see 6.1).

5.2.1 Unit of issue. One bottle containing 100 tablets, as specified, constitutes one unit of issue.

5.2.2 Level A.

5.2.2.1 Immediate container. The immediate container shall be a commercially available immediate bottle considered acceptable by the Food and Drug Administration (FDA) for the product contained therein and shall be of such compatibility that neither the contents nor the container are altered in any way by each other.

5.2.2.2 Unit packaging. At the option of the contractor, each unit may be packaged as specified in 5.2.5 of PPP-C-186.

5.2.2.3 Intermediate packaging. Twelve (12) units of issue shall be packaged as specified in 5.2.7.1 of PPP-C-186.

5.2.3 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.3 Packing. Packing shall be level A, B or C, as specified (see 6.1).

5.3.1 Level A. Two hundred eighty-eight (288) units of issue shall be packed as specified in 5.4.4 of PPP-C-186.

5.3.2 Level B. Two hundred eighty-eight (288) units of issue shall be packed as specified in 5.4.3.1.1 of PPP-C-186.

5.3.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.4 Marking.

5.4.1 Labeling. Labeling shall be in accordance with the requirements of the Federal Food, Drug and Cosmetic Act and shall include the information required below:

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5.4.1.1 Immediate container. Each immediate container label shall bear the following information:

(a) label information in accordance with commercial practice.

(b) the expiration date.

5.4.1.2 Unit package. Each unit package label shall bear the following information:

(a) label information in accordance with commercial practice.

(b) the National Stock Number.

NOTES: (1) The National Stock Number shall not be preceded by any prefix.

(2) Strip labeling may be used.

(3) Label adhesion test shall not apply.

(4) If a unit package is not supplied, the National Stock Number shall appear on the immediate container label.

5.4.2 Intermediate package. Each intermediate package shall be marked as specified in 5.5.3 of PPP-C-186. In addition, Type I shelf life marking as specified in MIL-STD-129 shall be shown. Marking shall include the legend:

"STORE AT CONTROLLED ROOM TEMPERATURE 15° - 30° C (59° - 86° F)"

5.4.3 Exterior container. Exterior containers shall be marked as specified in 5.5.4 of PPP-C-186. Type I shelf life marking as specified in MIL-STD-129 shall be shown. Marking shall include the legend:

"STORE AT CONTROLLED ROOM TEMPERATURE 15° - 30° C (59° - 86° F)"

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.

(b) National Stock Number.

(c) Selection of applicable levels of packaging and packing (see 5.2 and 5.3)

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6.2 This specification covers the following item:

National Stock Number	Item Identification
6505-01-017-1626	DIHYDROERGOCORNINE, DIHYDROERGOCRISTINE, AND DIHYDROERGOKRYPTINE MESYLATES TABLETS 1 mg, 100s

Custodians:
Army - MD
Navy - MS
Air Force - 03

Review activity:
DoD - MB

Preparing activity:
DLA-DM
Project No. 6505-1702

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

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DOCUMENT IDENTIFIER AND TITLE MIL-D-37555 DIHYDROERGOCORNINE, DIHYDROERGOCRISTINE, AND DIHYDROERGOKRYPTINE MESYLATES TABLETS

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

DD FORM 1 JAN 72 1426 REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED

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